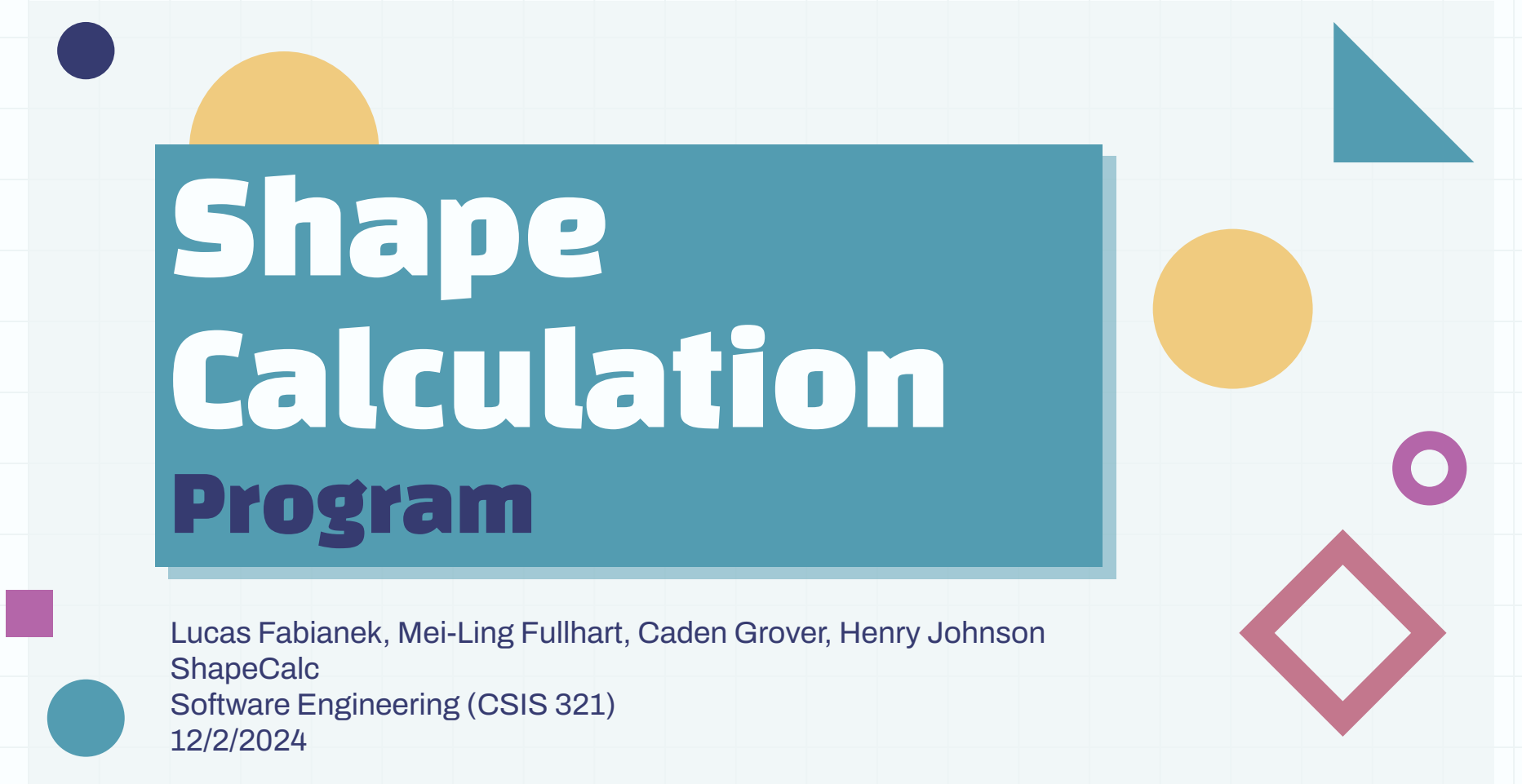




Shape Calculation Program

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ShapeCalc
Software Engineering (CSIS 321)
12/2/2024

Overview

Brief Introduction

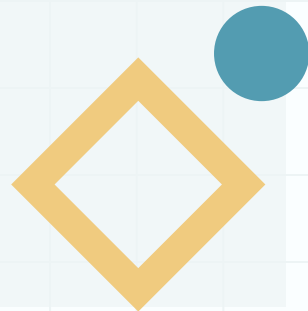
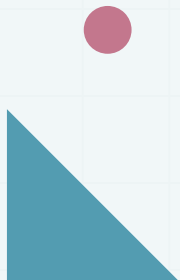
The Shape Calculation Program is a command-line program to do simple geometric calculations on:

- Rectangle
- Square
- Circle
- Right Triangle

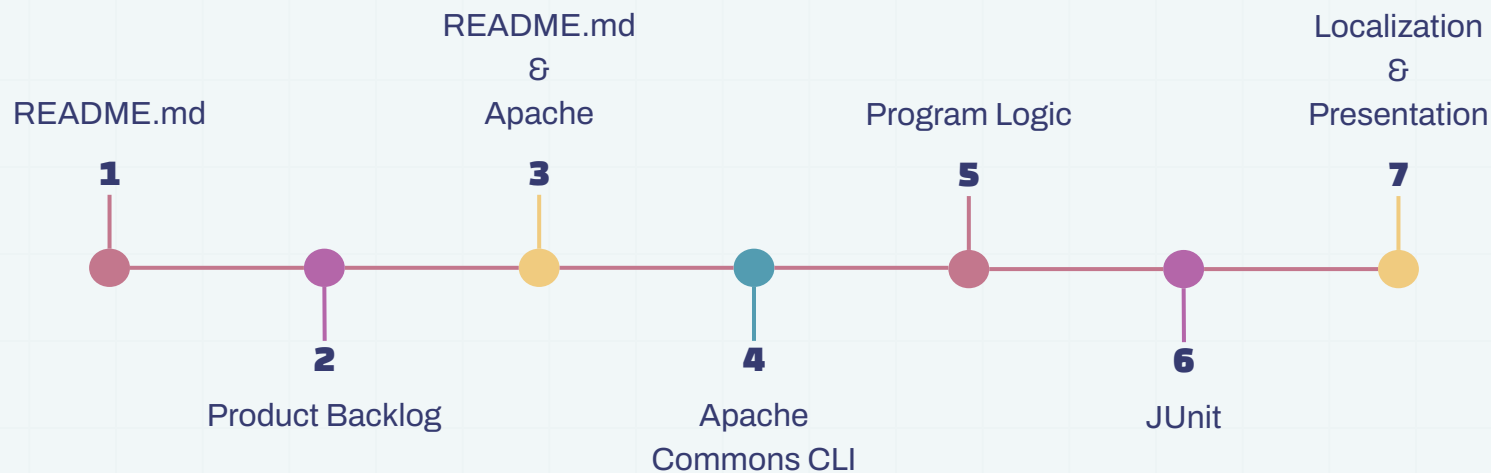
Key Goals & Objectives

The main goal of the Shape Calculation program is to provide an easy-to-use tool to calculate:

- Perimeter
- Area



Sprints Framework



Sprint 1

Overview & High-Level Features

Goals

Take various types of shapes & performs various mathematical features with the given shape

Key Tasks

Functional Requirements
Non-Functional Requirements
Time Estimate
Description

Outcome

Full list of Functional & Non-Functional Requirements

Functional Requirements:

Functional Requirements	Time Estimate	Description
Perimeter Calculation	30 Mins	Uses the correct formula for the given shape to produce the the given perimeter of the shape
Area Calculation	30 Mins	Uses the correct formula for the given shape to produce the given area of the shape
Compare Shapes	15 Mins	Takes the perimeter or the area to show which shape is bigger or smaller
Volume Calculation	30 Mins	Takes the given components for the shape and returns the the volume of that shape
Unit Conversion	30 Mins	Be able to take various measurement types such as imperial and metric and be able to perform the correct calculations with those values
Multiple Shape Support	45 Mins	Takes all sorts of shapes and uses the correct formulas for each shape type
Error Handling	30 Mins	Double checks the users inputs to make sure that the correct values are given

Sprint 2

Detailed Product Backlog

Goals

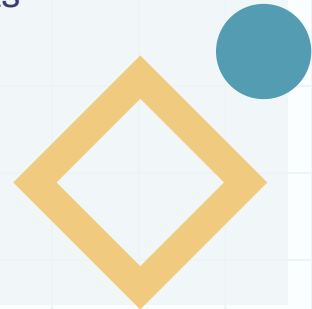
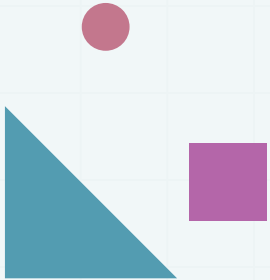
Build out a complete & detailed Product Backlog

Key Tasks

Create issues in GitLab & assign them to the product Backlog

Outcome

Complete Product Backlog initialized, issues assigned to respective sprints



Sprint 3

Outline Command-Line Options

Goals

Use Apache Commons CLI to write code to parse command line arguments

Key Tasks

Learn Apache Commons CLI

Outcome

Apache Commons CLI learned, began developing command-line arguments

Example of Apache Commons

```
-- shapetype -- circle -- radius -- 2.5 -- centimeters -- area
```

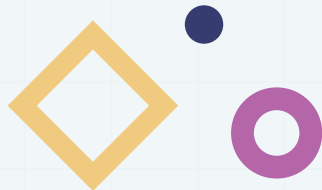
print the output

```
-- shapetype -- rectangle -- length -- 4.5 -- centimeters -- width -- 4.0 -- centimeters -- perimeter
```

print the output

```
-- shapetype -- right triangle -- base -- 2.5 -- centimeters -- height -- 3.0 -- centimeters -- area
```

print the output



Sprint 4

Apache Commons CLI

Goals

Complete Apache Commons CLI code
Began writing java code

Key Tasks

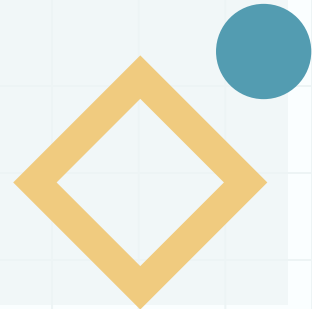
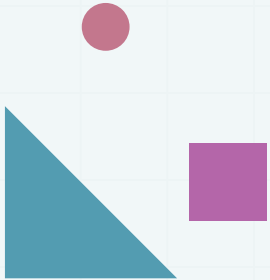
Implement Apache Commons CLI
Begin writing java code for testing

Outcome

Accepts command line arguments & displays the correct output for the given shape and function

Challenges

Correctly implementing Apache Commons



Sprint 5

Program Logic

Goals

Write shape class, and four unique shape classes

Key Tasks

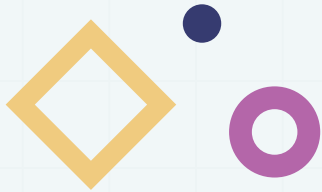
Write java code for shape
Write java code for circle, square, rectangle, & right triangle

Outcome

Functioning code that took command line arguments & returned shape calculation results

Challenges

Integrating the Apache Commons API & Java program logic



Sprint 6

JUnit Testing

Goals

Create tests using Junit in order to find any errors in our program

Key Tasks

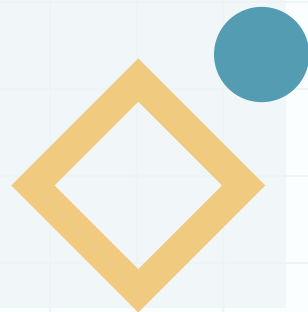
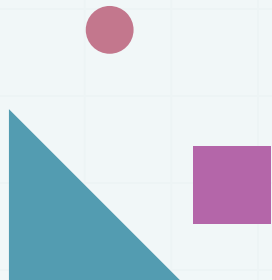
Setting up Junit in our IDE & writing the test cases

Outcome

Functioning Junit testers
Found many issues within our original code that were then fixed

Challenges

Time taken to complete fixes found using JUnit



Sprint 7

Localization & Presentation

Goals

Support EFIGS & complete the presentation

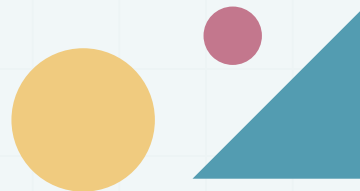
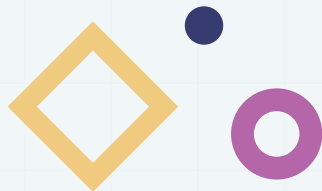
Key Tasks

Add translation support for 5 languages(EFIGS) & create the presentation

Outcome

Added support for French, Italian, German, & Spanish
Completed the presentation

```
# Help and general messages
missing_measure=Fehler: Argument 'measure' fehlt. Bitte geben Sie 'Fläche' oder 'Umfang' an.
unsupported_shape=Fehler: Nicht unterstützter Formtyp: {0}
missing_radius=Fehler: Argument 'Radius' fehlt für den Kreis.
missing_side=Fehler: Argument 'Seite' fehlt für das Quadrat.
missing_length_or_width=Fehler: Argument 'Länge' und/oder 'Breite' fehlt für das Rechteck.
missing_base_or_height=Fehler: Argument 'Basis' und/oder 'Höhe' fehlt für das rechtwinklige Dreieck.
```



Gantt Chart



Final Demo

Overview

Accepts command line arguments for 4 shapes with 2 different functions. The area and the perimeter. All functionality is supported through EFIGS.

Highlights

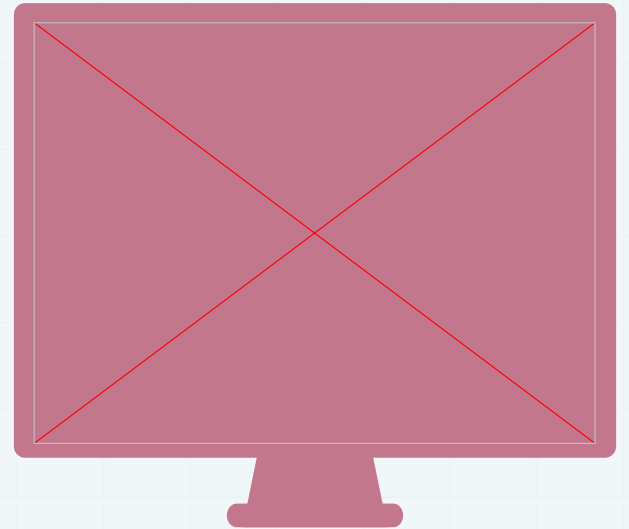
Example:

Input: `java Apache -lang fr -ctry FR -st circle -ra 5 -m area`

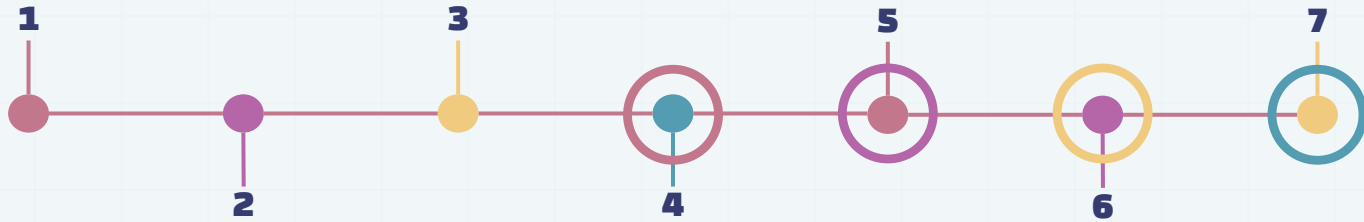
Output:

Shape Type: circle

La superficie de la circle
est de 78.54.



Testing



Tools & Methods

Automated: JUnit
Manual: CLI
Frameworks: JUnit &
Apache Commons CLI

Key Results

Test Coverage: 90%
Bugs: Boundary Conditions,
CLI Parsing

Challenges & Learning

Team

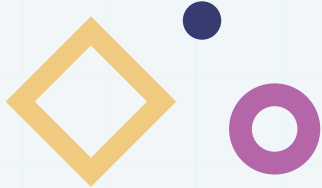
- Time management
- Workload Distribution
- Gitlab

Technical

- Apache Commons CLI
- Apache Commons CLI & Java Program Logic
- Localization & Apache

Lessons Learned

- Apache Commons CLI
- Localization
- Integration



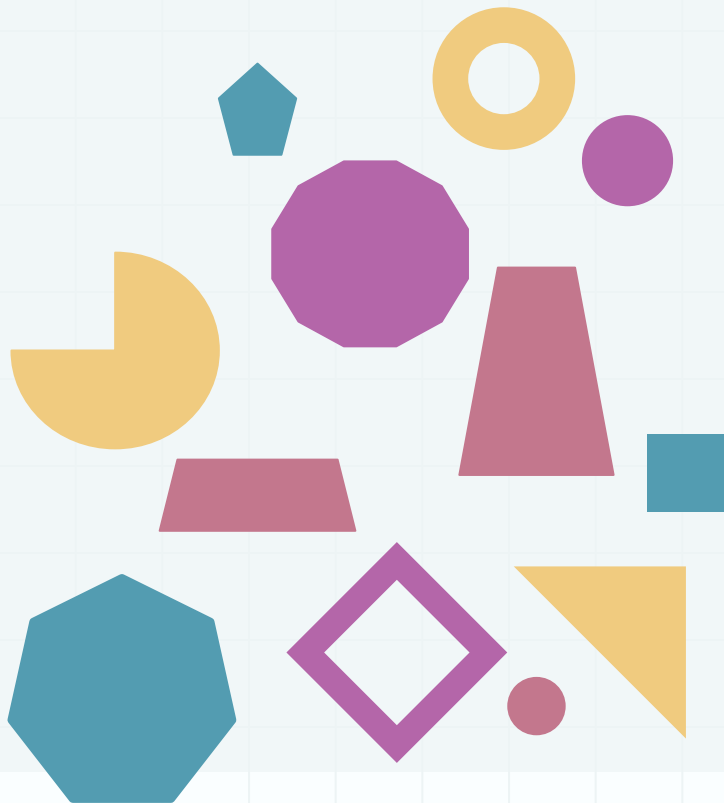
Future Work

Pending Features

- Backwards Functionality
 - Radius, Circumference, Length, Width, etc
- Cloud Saving
- Visual Display
- More Shapes

Improvements

- Better time management
- Git issues log maintenance
- Streamline Apache Commons CLI
- JUnit
- Maintainability



Conclusion

Functioning program that accepts command line arguments for 4 shapes with 2 different functions, area and perimeter. All functionality is supported through EFIGS.

Acknowledgements

Brian Snider

Thank You

